ESE 4115 One-Shot Notes

Power Plant Engineering · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

One-shot notes: ESE 4115 Power Plant Engineering

Load curve words

Load curve = load vs time (in order). **Load duration curve (LDC)** = same loads sorted high → low. Area = energy (MWh). Ideal curve flat (LF = 1); realized curve has peaks.

Base load = always there (coal, nuclear, big hydro, combined cycle). **Peak load** = few hours (gas turbine, diesel, pumped storage).

Factors (A-M-I rule)

A = average load, M = maximum demand, I = installed capacity.

Load factor = A/M · capacity factor = A/I · utilisation = M/I · demand factor = M / connected load · diversity = Σ individual maxima / system max (**> 1**, the only one) · use factor = E / (I × hours run) · reserve = I − M.

Hours in service = 8760 × CF / use factor.

Cost per kWh recipe

1 crore = 10^7, 1 lac = 10^5. kWh/yr = kW × LF (or CF) × 8760. Fixed = (interest + depreciation)% × capital + fixed wages. Running = fuel Tk/kWh × kWh. Cost = total / kWh sent out (minus auxiliaries).

Answers: 2023 Tk 1.28 · 2020 Tk 1.06 · 2021 Tk 2.00 per kWh.

Economics words

Fixed cost: interest, depreciation, insurance, taxes, salaries. **Running**: fuel, oil, water, maintenance, output wages.

Depreciation = loss of value; causes WADOI (wear, age, deterioration, obsolescence, inadequacy). Methods: straight line $(P - S)/n$, diminishing value, sinking fund $q = (P - S)r/[(1 + r)^n - 1]$.

Economic load sharing: peak plant runs $(R_1 - R_2)/(P_2 - P_1)$ h/yr (extra fixed per kW over extra running per kWh).

Tariff

Y = DX + EZ + C (demand × kW + energy × kWh + constant). Flat demand Y = DX; straight-line meter Y = EZ; step meter (whole use at one step rate, rate falls); block rate $\Sigma E_i Z_i$; two-part (Hopkinson) DX + EZ; three-part (Doherty) DX + EZ + C; time-of-use.

Objectives: recover generation, transmission/distribution, operation costs + profit. Requirements: fair, simple, reasonable profit, encourage off-peak use and high power factor.

Load forecasting

Purposes: plan capacity, unit commitment, fuel, maintenance, reserve, tariff. Factors TWEDS-R: time, weather, economy, demography, special events, random.

Short (hour-week), medium (month-year), long (> 1 yr). Methods: similar-day, regression, ARMA/ARIMA, ANN, expert system. Additive $L = L_n + L_w + L_s + L_r$.

Peak approach (extrapolate MW) vs energy approach (forecast kWh, peak = E/(LF × 8760)).

Bangladesh

Gas about 55 % of capacity (two-thirds of energy), furnace oil 27 %, import 9 %, coal 6 %, diesel 2 %, hydro 230 MW (Kaptai), renewables about 2.7 % (mostly solar home systems). Cost: hydro < gas < coal < oil < diesel.

PSMP 2015: peak 8,921 MW (2015) → about 50,400 MW (2041), need about 57,000 MW with reserve, GDP 6 %/yr. Meet with imported coal (Payra, Rampal, Matarbari 1200-1320 MW each), LNG + gas combined cycle, nuclear Rooppur 2 × 1200 MW (Russia), imports (Bheramara, Adani), renewables, efficiency.

Variable load and siting

Effects: extra equipment, higher production cost, reserve capacity, low LF, part-load efficiency loss, peak plants needed, control/load prediction problems.

Site: fuel, water, load centre, land, transport, ash disposal, environment, labour, expansion. Grid benefits: less reserve, cheapest plants first, reliability, diversity, big units, power exchange.

Hydro plant

Catchment → reservoir → dam (+ spillway) → gate, tunnel → surge tank → penstock → turbine-generator → draft tube → tail race.

Surge tank: absorbs water hammer on load drop, supplies water on load rise. **Forebay**: basin at penstock head, temporary storage. **Spillway**: releases flood water. Dams: gravity (weight), arch (arch action, gorges), buttress (slab on buttresses), earth.

Hydrograph Q vs time; flow duration curve Q vs % time; mass curve cumulative volume vs time; run-off = rain reaching streams.

Hydro numbers and turbines

Q = area × rain × fraction / 31.536 × 10^6 s. **P (kW) = 9.81 Q H η**. Capacity = P / LF. Energy (J) = m g H η; 1 kWh = 3.6 MJ.

Answers: 2023 6.67 MW (13.3 MW capacity) · 2020 6.58 MW (13.2) · 2022 3.36 MW (7.47) · 2021 24.6 MW · 2024 3,567 GWh/yr.

Pelton (impulse) high head; Francis medium; Kaplan low (< 50 m). $N_s = N\sqrt{P}/H^{5/4}$; N = 120f/p. Governor: fly balls → relay valve → servomotor → spear (+ deflector).

Diesel plant

Systems FACLSG: fuel, air intake/exhaust, cooling, lubrication, starting, governing. Fuel path: storage → day tank → filter → injection pump → injector. Solid injection (no air compressor) beats air-blast.

Uses: peak, standby, mobile, small central/island, nursery and starting stations. Peak-load: starts in minutes, no standby loss, low capital.

Cooling needed: walls < 400 °C, oil film, valves, ηv. Thermostat keeps water in jacket until 70-85 °C then opens to radiator. Starting: compressed air (big), electric (small), auxiliary engine.

Gas turbine

Open: air → compressor → combustor → turbine → exhaust. Closed: heater + cooler, same fluid.

$x = r_p^{0.2857}$; T2s = T1x; T4s = T3/x; $T2 = T1 + (T2s - T1)/\eta_c$; $T4 = T3 - \eta_t(T3 - T4s)$; $w = 1.005\,\Delta T$; $\eta = w_{net}/q_{in}$; ideal $\eta = 1 - 1/x$; $bwr = w_c/w_t$. Regenerator $T5 = T2 + \varepsilon(T4 - T2)$.

Max work: $T2 = T4 = \sqrt{(T1T3)}$. Regeneration: η ↑, work same (needs T4 > T2). Intercooling: w_c ↓.

Reheat: w_t ↑. All three + regenerator: both ↑ (Ericsson).

Answers: 2020 46.6 % · 2021 27.4 % → 35.6 % · 2022 37.5 % → 68.9 %.

Combined, reheat, binary

Combined: GT exhaust → HRSG → steam turbine. $\eta = \eta_{GT} + \eta_{ST}(1 - \eta_{GT}) \approx$ 50-62 %. Gas on top, steam below.

Reheat Rankine: HP turbine → reheater → LP turbine: drier exhaust, more work, η +4-5 %.

Binary: mercury cycle on top, its condenser boils water. Not used: costly, toxic, does not wet metal.

Exergy (2024)

$X_{destroyed} = T0\,s_{gen}$. $X_{in} = q_{in}(1 - T0/T_s)$. $\eta_{II} = w_{net}/X_{in}$. Check $X_{in} - \Sigma X_d = w_{net}$.

Diesel 5(c): T2 1094.9 K, P2 5424 kPa, T4 785.5 K, w_{net} 299.1 kJ/kg, η 47.4 %, s_{gen} 0.791, X_d 237.4, η_{II} 55.7 %.

Brayton 6(a): T2 709 K, T4 1058 K, W_{net} 4.66 MW, η 36.4 %, s_{gen} 22.65 kW/K, X_d 6.64 MW, η_{II} 41.3 %.

Steam plant

Pump → boiler → turbine → condenser. Circuits CAFe Cool: coal-ash, air-flue gas, feed water-steam, cooling water. Base-load: big units, cheap fuel, long life, steady efficiency.

Design activities: load survey, site, fuel, capacity and units, main equipment, auxiliaries, layout, economics, EIA.

Feed water

Impurities: hardness salts (scale), O_2/CO_2 (corrosion), suspended solids (sludge), oil/alkali (foaming, priming, embrittlement).

Treat: mechanical (settle, coagulate, filter), thermal (deaerate, distil), chemical (lime-soda, zeolite), demineralise, blow-down. Zeolite: Na_2Z + Ca salt → CaZ + Na salt; regenerate with brine.

Draught, chimney, cooling tower

Natural (chimney, 10-20 mm water) vs mechanical: forced (FD fan at inlet), induced (ID fan near chimney), balanced (both, furnace slightly below atmospheric).

$h_w = 353\ H\ [1/T_a - ((m + 1)/m)/T_g]$ mm water; $H_g = H[(m/(m + 1))T_g/T_a - 1]$. 1 mm water = 1 kgf/m^2.

2023: 15.2 mm, 23.9 m; 2020: H = 43.6 m.

Range = $T_{in} - T_{out}$; approach = T_{out} − WBT; efficiency = range/(range + approach). 2021: T_{out} 22.4 °C, evaporated 1.03 %, range 7.6, approach 7.2 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AC	alternating current
ANN	artificial neural network
ARIMA	auto-regressive integrated moving average
ARMA	auto-regressive moving average
BHP	brake horsepower
BIFPCL	Bangladesh-India Friendship Power Company Ltd
BPDB	Bangladesh Power Development Board
CCGT	combined cycle gas turbine
CCPP	combined cycle power plant
CF	capacity factor
CHP	combined heat and power (cogeneration)
CI	compression ignition (diesel)
CPGCBL	Coal Power Generation Company Bangladesh Ltd
CV	calorific value (heat released by 1 kg of fuel)
DBT	dry bulb temperature
EGCB	Electricity Generation Company of Bangladesh
ESP	electrostatic precipitator
FBC	fluidised bed combustion
FD	forced draught
GT	gas turbine
GWh	gigawatt-hour = 10^6 kWh
HCV	higher calorific value
HFO	heavy fuel oil
HP	high pressure
HRSG	heat recovery steam generator
HSD	high speed diesel
ID	induced draught
IP	intermediate pressure
IPP	independent power producer
kW	kilowatt = 1000 W
kWh	kilowatt-hour = 1 unit of electrical energy
LDC	load duration curve
LF	load factor
LNG	liquefied natural gas
LP	low pressure
LTLF	long-term load forecasting

MTLF	medium-term load forecasting
MW	megawatt = 1000 kW
MWh	megawatt-hour = 1000 kWh
NOx	nitrogen oxides
NWPGCL	North-West Power Generation Company Ltd
PSMP	Power System Master Plan
PV	photovoltaic (solar cells)
RE	renewable energy
RH	relative humidity
RPCL	Rural Power Company Ltd
rpm	revolutions per minute
SI	spark ignition (petrol)
SOx	sulphur oxides
ST	steam turbine
STLF	short-term load forecasting
TDC	top dead centre
Tk	Taka (Bangladeshi currency)
ToU	time of use
TWL	tail water level
UF	use (utilisation) factor
WBT	wet bulb temperature